Mohammad Robati Shirzad

https://mohrobati.github.io
https://www.linkedin.com/in/moh-robati/

Email : moh.robati.shirzad@gmail.com
Mobile : +1-226-698-3672
Address: Toronto, Ontario, Canada

Education / Training

- **AWS Certified Solutions Architect — Associate** — AWS
  Amazon Web Servers (link) — *Dec. 2022 – Dec. 2025*
- **Master of Applied Science in Computer Software** — Waterloo, ON, Canada
  University of Waterloo, Electircal and Computer Engineering Dept. — *Sep. 2021 – May 2023*
- **Bachelor of Science in Computer Engineering** — Tehran, Iran
  Tehran Polytechnic — *Sep. 2017 – Sep. 2021*

Experience

- **Senior Software Engineer** — Toronto, ON
  Huawei Canada (contractor full-time until Jan 2024, then permanent full-time) — *May 2023 – Present*
  - Led end-to-end design, development, and deployment of large-scale multi-agent AI systems on cloud platforms, building evaluation pipelines to benchmark agent behavior, reasoning quality, coordination efficiency, and task execution performance across complex environments. Designed and implemented agent coordination strategies, including reinforcement learning-based policies, hierarchical planning architectures, and emergent communication protocols to enable robust multi-agent collaboration and autonomous decision-making. Architected and scaled GenAI-driven systems for intelligent collaboration, planning, and execution, integrating cutting-edge research into production-grade agent orchestration frameworks and cloud services. Collaborated with global AI researchers and engineering teams to translate emerging agentic AI methodologies into system implementations, continuously evolving capabilities through integration of new models, orchestration techniques, and evaluation methods.
  - Working closely with the research and development team to deliver our high quality, secure WASM based Cloud Services product. Also, helping with expand the capabilities of the product with new functionality and create a robust WASM based serverless platform with the latest technologies. Quickly onboarded, and started contributing within just two weeks of joining. Helped with a lot of issues involving adding new features, addressing security vulnerabilities, and executing codebase refactoring. Based on the feedback, after joining the team, my contributions accelerated the development timelines.
- **Lab Instructor / Teaching Assistant** — Waterloo, ON
  University of Waterloo — *Jan. 2022 – April 2023*
  - As a TA, I instructed high-performance, multithreaded programming in Rust and C, conducting code review sessions that significantly improved students' coding skills, leading to many securing positions at major tech companies. I streamlined professor tasks by developing automated grading scripts, ensuring efficient feedback delivery for the students. I received excellent reviews throughout my tenure.

Skills

- **Programming Languages**: Rust, Javascript, Python, C/C++, Java, Racket
  Frameworks: PyTorch, FSDP, MindSpore, Ray, CUDA, LLVM, Rust async runtime (Tokio), WebAssembly (WASM), distributed systems frameworks, agentic AI frameworks, LLM fine-tuning stacks
  Cloud & Distributed Systems: AWS (EC2, S3, RDS, Aurora, DynamoDB, Lambda, SNS, SQS, CloudFront), Huawei Cloud (ECS, OBS, RDS, FunctionGraph, ModelArts), Kubernetes, Docker, distributed training pipelines, microservices, serverless architectures, cloud-native systems
  AI / ML Systems: Multi-agent systems, agent orchestration, AI agent evaluation, reinforcement learning (RLHF-adjacent workflows), SFT, DPO, GRPO, model fine-tuning, data pipelines, data augmentation, training scalability, LLM systems design, prompt engineering, context-window engineering
  Systems & Low-Level Engineering: Virtual Machines, sandbox isolation, LLVM toolchains, OS-level abstractions, low-latency systems, concurrency, memory safety, performance optimization
  Accelerators & Parallelism: GPU/accelerator computing (NVIDIA H100, Ascend NPUs), CUDA kernels, 3D parallelization, distributed training, model parallelism, pipeline parallelism, tensor parallelism
  Engineering Practices: CI/CD (Jenkins), Git, Agile/Scrum, production system design, observability, debugging distributed systems, scalability engineering
  Languages: English, French, Persian